

Lesson 11: Confidence Intervals Continued + Review

BSTA 551: Statistical Inference

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What We've Covered (Lessons 7-10)

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Today's Goals:

1. Review the three main CI construction approaches
2. Extend Wald intervals to other distributions and functions of MLEs
3. Learn the **delta method** for variance of transformed estimators
4. See score intervals for non-proportion parameters

The Big Picture

Three Approaches to CI Construction

Three Main Approaches to Confidence Interval Construction



**Pivotal
Quantity**

Exact distribution
known (z , t , χ^2)



**Asymptotic
(Wald, Score)**

Large sample
normality of MLE and Score



Bootstrap

Resample from
data itself

Quick Reference: CI Formulas

Parameter	Method	Formula
μ (σ known)	Pivotal (z)	$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$
μ (σ unknown)	Pivotal (t)	$\bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$
p (proportion)	Wald	$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
p (proportion)	Score	$\frac{\hat{p} + \frac{z^2}{2n} \pm z \sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + z^2/n}$
λ (Poisson)	Score	$\frac{X + z^2/2 \pm z \sqrt{X + z^2/4}}{n}$
Any θ	Bootstrap Percentile	$(\hat{\theta}_{\alpha/2}^*, \hat{\theta}_{1-\alpha/2}^*)$
Any θ	Simple Bootstrap t	$\hat{\theta} \pm t_{\alpha/2, n-1} \cdot SE_{boot}$
Any θ	Studentized Bootstrap t	$(\hat{\theta} - t_{1-\alpha/2}^* \cdot SE_{\hat{\theta}}, \hat{\theta} - t_{\alpha/2}^* \cdot SE_{\hat{\theta}})$

Correct CI Interpretation

Valid Interpretations

- “We are 95% confident that the true parameter lies in (L, U)”
- “The interval (L, U) is a plausible range for the parameter”
- “In repeated sampling, 95% of such intervals contain the true parameter”

Invalid Interpretations

- “There is a 95% probability that θ is in (L, U)” ✗
- “95% of the data falls between L and U” ✗
- “The true parameter varies between L and U” ✗

Wald Intervals: General Framework

The Wald Interval Principle

! Asymptotic Normality of MLEs

Under regularity conditions, the MLE $\hat{\theta}$ is asymptotically normal:

$$\hat{\theta} \overset{approx}{\sim} N\left(\theta, \frac{1}{I_n(\theta)}\right)$$

where $I_n(\theta) = nI_1(\theta)$ is the **total Fisher information** for n observations, and $I_1(\theta)$ is the Fisher information for a single observation.

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The Wald CI plugs in $\hat{\theta}$ for unknown quantities in the variance:

$$\hat{\theta} \pm z_{\alpha/2} \cdot \widehat{SE}(\hat{\theta})$$

where $\widehat{SE}(\hat{\theta}) = \sqrt{1/I_n(\hat{\theta})}$ or the observed information.

Wald Interval: Poisson Rate

Setting: Count data $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Poisson}(\lambda)$

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MLE: $\hat{\lambda} = \bar{X}$

Fisher Information: $I_1(\lambda) = 1/\lambda$, so $I_n(\lambda) = n/\lambda$

Asymptotic Variance: $\text{Var}(\hat{\lambda}) \approx 1/I_n(\lambda) = \lambda/n$

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 Wald CI for Poisson Rate

$$\hat{\lambda} \pm z_{\alpha/2} \sqrt{\frac{\hat{\lambda}}{n}}$$

Medical Example: Hospital Infection Rate

A hospital monitors central line-associated bloodstream infections (CLABSI). Over $n = 500$ patient-days, they observe $X = 8$ infections.

```
1 x <- 8          # observed infections
2 n <- 500        # patient-days
3 lambda_hat <- x / n # MLE (rate per patient-day)
4
5 # Wald CI
6 se_wald <- sqrt(lambda_hat / n)
7 z <- qnorm(0.975)
8 wald_ci <- lambda_hat + c(-1, 1) * z * se_wald
9
10 cat("MLE (rate per 1000 patient-days):", round(lambda_hat * 1000, 2), "\n")
```

MLE (rate per 1000 patient-days): 16

```
1 cat("95% Wald CI:", round(wald_ci * 1000, 3), "per 1000 patient-days")
```

95% Wald CI: 4.913 27.087 per 1000 patient-days

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MLE (rate per 1000 patient-days): 16

```
1 cat("95% Wald CI:", round(wald_ci * 1000, 3), "per 1000 patient-days")
```

95% Wald CI: 4.913 27.087 per 1000 patient-days

Note: The lower bound is positive, but for very small counts, Wald can give negative bounds!

Wald Interval: Exponential Rate

Setting: Failure times $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Exp}(\lambda)$

MLE: $\hat{\lambda} = 1/\bar{X}$

Fisher Information: $I_1(\lambda) = 1/\lambda^2$, so $I_n(\lambda) = n/\lambda^2$

Asymptotic Variance: $\text{Var}(\hat{\lambda}) \approx 1/I_n(\lambda) = \lambda^2/n$

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 Wald CI for Exponential Rate

$$\hat{\lambda} \pm z_{\alpha/2} \frac{\hat{\lambda}}{\sqrt{n}}$$

Medical Example: Time to Symptom Relief

A clinical trial measures time (hours) until symptom relief for $n = 30$ patients taking a new analgesic:

```
1 # Simulated relief times (hours)
2 set.seed(551)
3 relief_times <- rexp(30, rate = 0.5) # True rate = 0.5 (mean = 2 hours)
4
5 n <- length(relief_times)
6 lambda_hat <- 1 / mean(relief_times)
7
8 # Wald CI for rate
9 se_lambda <- lambda_hat / sqrt(n)
10 wald_ci_lambda <- lambda_hat + c(-1, 1) * qnorm(0.975) * se_lambda
11
12 cat("Sample mean relief time:", round(mean(relief_times), 2), "hours\n")
```

Sample mean relief time: 2.01 hours

```
1 cat("MLE of rate:", round(lambda_hat, 3), "per hour\n")
```

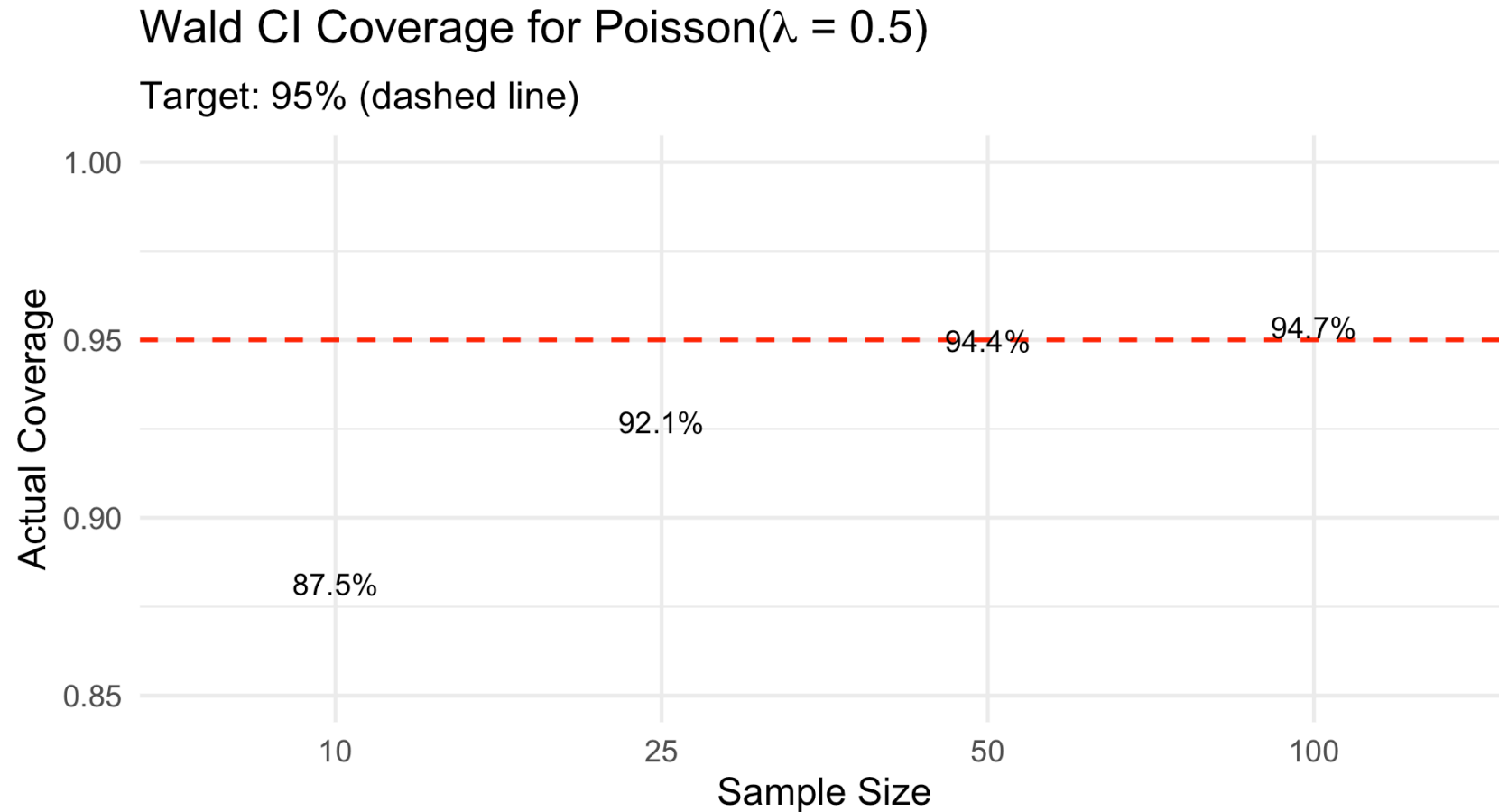
MLE of rate: 0.498 per hour

```
1 cat("95% Wald CI for rate:", round(wald_ci_lambda, 3), "per hour")
```

95% Wald CI for rate: 0.32 0.677 per hour

When Wald Intervals Struggle

► Code



Takeaway: Wald intervals can have poor coverage for small samples or extreme parameters.

The Delta Method

Motivation: Functions of MLEs

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Examples:

- Mean survival time $\mu = 1/\lambda$ when we estimate $\hat{\lambda}$
- Odds $\omega = p/(1 - p)$ when we estimate $\hat{p}$
- Log-odds $\eta = \log(p/(1 - p))$ for logistic regression

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Question: If $\hat{\theta} \overset{approx}{\sim} N(\theta, \sigma_{\hat{\theta}}^2)$, what is the distribution of $g(\hat{\theta})$?

The Delta Method

! Delta Method (First-Order Approximation)

If $\hat{\theta} \overset{approx}{\sim} N(\theta, \sigma_{\hat{\theta}}^2)$ and g is differentiable at θ with $g'(\theta) \neq 0$, then:

$$g(\hat{\theta}) \overset{approx}{\sim} N\left(g(\theta), [g'(\theta)]^2 \cdot \sigma_{\hat{\theta}}^2\right)$$

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$$g(\hat{\theta}) \overset{approx}{\sim} N\left(g(\theta), [g'(\theta)]^2 \cdot \sigma_{\hat{\theta}}^2\right)$$

Intuition: Taylor expansion around θ :

$$g(\hat{\theta}) \approx g(\theta) + g'(\theta)(\hat{\theta} - \theta)$$

$$\text{So } \text{Var}[g(\hat{\theta})] \approx [g'(\theta)]^2 \cdot \text{Var}(\hat{\theta})$$

Delta Method: Variance Formula

Practical Formula

$$\widehat{SE}[g(\hat{\theta})] = |g'(\hat{\theta})| \cdot \widehat{SE}(\hat{\theta})$$

Wald CI for $g(\theta)$:

$$g(\hat{\theta}) \pm z_{\alpha/2} \cdot |g'(\hat{\theta})| \cdot \widehat{SE}(\hat{\theta})$$

Example: CI for Mean Survival Time

Setting: $X_1, \dots, X_n \sim \text{Exp}(\lambda)$, want CI for mean $\mu = 1/\lambda = g(\lambda)$

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Step 3: Apply delta method:

$$SE(\hat{\mu}) = |g'(\hat{\lambda})| \cdot SE(\hat{\lambda}) = \frac{1}{\hat{\lambda}^2} \cdot \frac{\hat{\lambda}}{\sqrt{n}} = \frac{1}{\hat{\lambda}\sqrt{n}} = \frac{\bar{X}}{\sqrt{n}}$$

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This is just $SE(\bar{X}) = s/\sqrt{n}$ when $s \approx \bar{X}$ for exponential data!

Medical Example: Mean Time to Relief (continued)

```
1 # From earlier: relief_times data
2 xbar <- mean(relief_times)
3 n <- length(relief_times)
4 lambda_hat <- 1 / xbar
5
6 # Delta method SE for mean (mu = 1/lambda)
7 # g'(lambda) = -1/lambda^2, so SE(mu) = SE(lambda)/lambda^2 = 1/(lambda * sqrt(n))
8 se_mu_delta <- 1 / (lambda_hat * sqrt(n))
9
10 # Equivalently: se_mu_delta = xbar / sqrt(n)
11 cat("Delta method SE for mean:", round(se_mu_delta, 3), "\n")
```

Delta method SE for mean: 0.366

```
1 cat("Direct SE (s/sqrt(n)):", round(sd(relief_times)/sqrt(n), 3), "\n\n")
```

Direct SE (s/sqrt(n)): 0.287

```
1 # 95% CI for mean survival time
2 ci_mu <- xbar + c(-1, 1) * qnorm(0.975) * se_mu_delta
3 cat("95% CI for mean relief time:", round(ci_mu, 2), "hours")
```

95% CI for mean relief time: 1.29 2.72 hours

Example: CI for Odds from Proportion

Setting: $X \sim \text{Binomial}(n, p)$, want CI for odds $\omega = p/(1 - p)$

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Step 1: We have $\hat{p} = X/n$ with $SE(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

Step 2: Define $g(p) = \frac{p}{1-p}$ and compute the derivative:

$$g'(p) = \frac{(1-p) \cdot 1 - p \cdot (-1)}{(1-p)^2} = \frac{1-p+p}{(1-p)^2} = \frac{1}{(1-p)^2}$$

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Step 3: Apply delta method:

$$SE(\hat{\omega}) = |g'(\hat{p})| \cdot SE(\hat{p}) = \frac{1}{(1-\hat{p})^2} \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \frac{\sqrt{\hat{p}(1-\hat{p})}}{(1-\hat{p})^2 \sqrt{n}}$$

Example: CI for Odds (Calculation)

```
1 # Clinical trial: 45 of 120 patients respond to treatment
2 x <- 45; n <- 120
3 p_hat <- x / n
4 omega_hat <- p_hat / (1 - p_hat) # Estimated odds
5
6 # Step-by-step delta method calculation
7 se_p <- sqrt(p_hat * (1 - p_hat) / n)
8 cat("SE(̂p) =", round(se_p, 4), "\n\n")
```

$SE(\hat{p}) = 0.0442$

```
1 g_prime <- 1 / (1 - p_hat)^2
2 cat("g'(\hat{p}) = 1/(1-\hat{p})^2 =", round(g_prime, 4), "\n\n")
```

$g'(\hat{p}) = 1/(1-\hat{p})^2 = 2.56$

```
1 se_omega <- abs(g_prime) * se_p
2 cat("SE(\hat{\omega}) = |g'(\hat{p})| \times SE(\hat{p}) =", round(g_prime, 4), "x", round(se_p, 4), "=", round(se_omega, 4))
```

$SE(\hat{\omega}) = |g'(\hat{p})| \times SE(\hat{p}) = 2.56 \times 0.0442 = 0.1131$

```
1 # 95% CI for odds
2 z <- qnorm(0.975)
3 ci_omega <- omega_hat + c(-1, 1) * z * se_omega
4
5 cat("Estimated odds:", round(omega_hat, 3), "\n")
```

Estimated odds: 0.6



Log Transformation: Often Better

For positive parameters (rates, odds), working on the **log scale** often improves coverage.

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Why? The sampling distribution of $\log(\hat{\omega})$ is often more symmetric than $\hat{\omega}$.

Step 1: Define $h(\omega) = \log(\omega)$, so $h'(\omega) = 1/\omega$

Step 2: Apply delta method to get SE of log-odds:

$$SE[\log(\hat{\omega})] = |h'(\hat{\omega})| \cdot SE(\hat{\omega}) = \frac{1}{\hat{\omega}} \cdot SE(\hat{\omega})$$

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Why? The sampling distribution of $\log(\hat{\omega})$ is often more symmetric than $\hat{\omega}$.

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Step 2: Apply delta method to get SE of log-odds:

$$SE[\log(\hat{\omega})] = |h'(\hat{\omega})| \cdot SE(\hat{\omega}) = \frac{1}{\hat{\omega}} \cdot SE(\hat{\omega})$$

Step 3: Build CI on log scale, then exponentiate back.

Log Transformation: Calculation

```
1 # Continuing from odds example
2 log_omega_hat <- log(omega_hat)
3
4 # Delta method for log(omega): h(omega) = log(omega), h'(omega) = 1/omega
5 h_prime <- 1 / omega_hat
6 se_log_omega <- abs(h_prime) * se_omega
7
8 cat("log(ω̂) =", round(log_omega_hat, 4), "\n")
```

$\log(\hat{\omega}) = -0.5108$

```
1 cat("h'(ω̂) = 1/ω̂ =", round(h_prime, 4), "\n")
```

$h'(\hat{\omega}) = 1/\hat{\omega} = 1.6667$

```
1 cat("SE[log(ω̂)] = |h'(ω̂)| × SE(ω̂) =", round(h_prime, 4), "×", round(se_omega, 4), "=",
2     round(se_log_omega, 4), "\n\n")
```

$SE[\log(\hat{\omega})] = |h'(\hat{\omega})| \times SE(\hat{\omega}) = 1.6667 \times 0.1131 = 0.1886$

```
1 # CI on log scale
2 ci_log_omega <- log_omega_hat + c(-1, 1) * qnorm(0.975) * se_log_omega
3 cat("95% CI for log(odds):", round(ci_log_omega, 3), "\n\n")
```

95% CI for log(odds): -0.88 -0.141

```
1 # Exponentiate to get CI for odds
2 ci_omega_transformed <- exp(ci_log_omega)
```



Log Transformation: Calculation

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1 # Continuing from odds example
2 log_omega_hat <- log(omega_hat)
3
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5 h_prime <- 1 / omega_hat
6 se_log_omega <- abs(h_prime) * se_omega
7
8 cat("log(ŵ) =", round(log_omega_hat, 4), "\n")
```

$\log(\hat{\omega}) = -0.5108$

```
1 cat("h'(ŵ) = 1/ŵ =", round(h_prime, 4), "\n")
```

$h'(\hat{\omega}) = 1/\hat{\omega} = 1.6667$

```
1 cat("SE[log(ŵ)] = |h'(ŵ)| × SE(ŵ) =", round(h_prime, 4), "×", round(se_omega, 4), "=",
2     round(se_log_omega, 4), "\n\n")
```

$SE[\log(\hat{\omega})] = |h'(\hat{\omega})| \times SE(\hat{\omega}) = 1.6667 \times 0.1131 = 0.1886$

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3 cat("95% CI for log(odds):", round(ci_log_omega, 3), "\n\n")
```

95% CI for log(odds): -0.88 -0.141

```
1 # Exponentiate to get CI for odds
2 ci_omega_transformed <- exp(ci_log_omega)
```



Delta Method: Summary

Common Delta Method Applications

Original Parameter	Function $g(\theta)$	Derivative $g'(\theta)$	$SE[g(\hat{\theta})]$
λ (Exp rate)	$\mu = 1/\lambda$	$-1/\lambda^2$	$1/(\hat{\lambda}\sqrt{n})$
p (proportion)	$\text{odds} = p/(1-p)$	$1/(1-p)^2$	$SE(\hat{p})/(1-\hat{p})^2$
p (proportion)	$\text{log-odds} = \log(p/(1-p))$	$1/(p(1-p))$	$1/\sqrt{n\hat{p}(1-\hat{p})}$
λ (Poisson)	$\log(\lambda)$	$1/\lambda$	$1/\sqrt{n\hat{\lambda}}$

Score Intervals: Beyond Proportions

Score Intervals: The General Principle

! Score Interval Derivation

The score interval inverts the **score test**. The score statistic is:

$$S(\theta) = \frac{[\ell'(\theta)]^2}{I_n(\theta)} \underset{\text{approx}}{\sim} \chi_1^2$$

The $100(1 - \alpha)\%$ score CI contains all θ such that $S(\theta) \leq z_{\alpha/2}^2$.

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The $100(1 - \alpha)\%$ score CI contains all θ such that $S(\theta) \leq z_{\alpha/2}^2$.

Equivalently: Find all θ where

$$\left| \frac{\ell'(\theta)}{\sqrt{I_n(\theta)}} \right| \leq z_{\alpha/2}$$

Understanding the Score Statistic

Components of the Score Statistic

$$S(\theta) = \frac{[\ell'(\theta)]^2}{I_n(\theta)} = \frac{(\text{Score Function})^2}{\text{Fisher Information}}$$

- **Numerator:** $\ell'(\theta) = \frac{\partial}{\partial \theta} \log L(\theta) = \sum_{i=1}^n \frac{\partial}{\partial \theta} \log f(X_i; \theta)$ is the **score function** – measures how much the log-likelihood is “pulling” away from θ
- **Denominator:** $I_n(\theta) = \text{Var}[\ell'(\theta)] = -E[\ell''(\theta)] = nI_1(\theta)$ is the **total Fisher information** – the variance of the score function (which equals n times the single-observation information)

Understanding the Score Statistic

Components of the Score Statistic

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- **Denominator:** $I_n(\theta) = \text{Var}[\ell'(\theta)] = -E[\ell''(\theta)] = nI_1(\theta)$ is the **total Fisher information** – the variance of the score function (which equals n times the single-observation information)

Intuition: At the true θ , the score function should be close to zero (since MLE maximizes likelihood). The score statistic measures how far $\ell'(\theta)$ is from zero, standardized by its variance.

Score vs Wald: Where the Variance is Evaluated

Aspect	Wald Interval	Score Interval
Test statistic	$\frac{\hat{\theta} - \theta}{\sqrt{I_n(\hat{\theta})^{-1}}}$	$\frac{\ell'(\theta)}{\sqrt{I_n(\theta)}}$
Variance evaluated at	$\hat{\theta}$ (the MLE)	θ (the hypothesized value)
Computation	Direct formula	Solve equation in θ
Coverage	Can be poor for extreme θ	Generally better
Bounds	May violate parameter space	Respects constraints

Score vs Wald: Where the Variance is Evaluated

Aspect	Wald Interval	Score Interval
Test statistic	$\frac{\hat{\theta} - \theta}{\sqrt{I_n(\hat{\theta})^{-1}}}$	$\frac{\ell'(\theta)}{\sqrt{I_n(\theta)}}$
Variance evaluated at	$\hat{\theta}$ (the MLE)	θ (the hypothesized value)
Computation	Direct formula	Solve equation in θ
Coverage	Can be poor for extreme θ	Generally better
Bounds	May violate parameter space	Respects constraints

Key insight: Score intervals evaluate variance at each candidate θ , not just at $\hat{\theta}$!

Score Interval for Proportion: Derivation

Setting: $X \sim \text{Binomial}(n, p), \hat{p} = X/n$

Score equation: Find p such that

$$\frac{(\hat{p} - p)^2}{p(1 - p)/n} \leq z_{\alpha/2}^2$$

Score Interval for Proportion: Derivation

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Score equation: Find p such that

$$\frac{(\hat{p} - p)^2}{p(1 - p)/n} \leq z_{\alpha/2}^2$$

Rearranging:

$$(\hat{p} - p)^2 \leq z^2 \cdot \frac{p(1 - p)}{n}$$

$$n(\hat{p} - p)^2 \leq z^2 p(1 - p)$$

$$n\hat{p}^2 - 2n\hat{p}p + np^2 \leq z^2 p - z^2 p^2$$

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$$n\hat{p}^2 - 2n\hat{p}p + np^2 \leq z^2 p - z^2 p^2$$

Collecting terms in p :

$$(n + z^2)p^2 - (2n\hat{p} + z^2)p + n\hat{p}^2 \leq 0$$

This is a quadratic in p – solve with the quadratic formula!



Score Interval for Proportion: Solution

The Wilson (Score) interval endpoints come from solving the quadratic:

$$p = \frac{2n\hat{p} + z^2 \pm \sqrt{z^4 + 4n\hat{p}z^2(1 - \hat{p})}}{2(n + z^2)}$$

Score Interval for Proportion: Solution

The Wilson (Score) interval endpoints come from solving the quadratic:

$$p = \frac{2n\hat{p} + z^2 \pm \sqrt{z^4 + 4n\hat{p}z^2(1 - \hat{p})}}{2(n + z^2)}$$

The **95% Score (Wilson) CI** for p is:

$$\left(\frac{\hat{p} + \frac{z^2}{2n} - z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}, \frac{\hat{p} + \frac{z^2}{2n} + z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}} \right)$$

```
1 # Compare Score vs Wald for extreme proportion
2 x <- 3; n <- 50; p_hat <- x/n
3 z <- qnorm(0.975)
4
5 # Wald
6 wald_ci <- p_hat + c(-1, 1) * z * sqrt(p_hat * (1 - p_hat) / n)
7
8 # Score (Wilson)
9 score_ci <- prop.test(x, n, correct = FALSE)$conf.int
```

Score Interval for Poisson: Derivation

Setting: $X \sim \text{Poisson}(n\lambda)$ (sum of n iid $\text{Poisson}(\lambda)$)

MLE: $\hat{\lambda} = X/n$

Variance under λ : $\text{Var}(\hat{\lambda}) = \lambda/n$

Score Interval for Poisson: Derivation

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MLE: $\hat{\lambda} = X/n$

Variance under λ : $\text{Var}(\hat{\lambda}) = \lambda/n$

Score equation: Find λ such that

$$\frac{(\hat{\lambda} - \lambda)^2}{\lambda/n} \leq z^2$$

Score Interval for Poisson: Derivation

Setting: $X \sim \text{Poisson}(n\lambda)$ (sum of n iid $\text{Poisson}(\lambda)$)

MLE: $\hat{\lambda} = X/n$

Variance under λ : $\text{Var}(\hat{\lambda}) = \lambda/n$

Score equation: Find λ such that

$$\frac{(\hat{\lambda} - \lambda)^2}{\lambda/n} \leq z^2$$

Rearranging:

$$n(\hat{\lambda} - \lambda)^2 \leq z^2 \lambda$$

$$n\hat{\lambda}^2 - 2n\hat{\lambda}\lambda + n\lambda^2 - z^2\lambda \leq 0$$

$$n\lambda^2 - (2n\hat{\lambda} + z^2)\lambda + n\hat{\lambda}^2 \leq 0$$

Score Interval for Poisson: Solution

Using the quadratic formula with $a = n$, $b = -(2n\hat{\lambda} + z^2)$, $c = n\hat{\lambda}^2$:

$$\lambda = \frac{(2n\hat{\lambda} + z^2) \pm \sqrt{(2n\hat{\lambda} + z^2)^2 - 4n^2\hat{\lambda}^2}}{2n}$$

Score Interval for Poisson: Solution

Using the quadratic formula with $a = n$, $b = -(2n\hat{\lambda} + z^2)$, $c = n\hat{\lambda}^2$:

$$\lambda = \frac{(2n\hat{\lambda} + z^2) \pm \sqrt{(2n\hat{\lambda} + z^2)^2 - 4n^2\hat{\lambda}^2}}{2n}$$

Simplifying, the **95% Score CI** for λ is:

$$\left(\frac{X + z^2/2 - z\sqrt{X + z^2/4}}{n}, \frac{X + z^2/2 + z\sqrt{X + z^2/4}}{n} \right)$$

Or in terms of $\hat{\lambda} = X/n$:

$$\left(\hat{\lambda} + \frac{z^2}{2n} - \frac{z}{n} \sqrt{n\hat{\lambda} + \frac{z^2}{4}}, \hat{\lambda} + \frac{z^2}{2n} + \frac{z}{n} \sqrt{n\hat{\lambda} + \frac{z^2}{4}} \right)$$

Medical Example: Hospital Infection Rate

```
1 # Hospital infection data (CLABSI)
2 x <- 8           # infections
3 n <- 500         # patient-days
4 z <- qnorm(0.975)
5
6 # Wald CI
7 lambda_hat <- x / n
8 wald_ci <- lambda_hat + c(-1, 1) * z * sqrt(lambda_hat / n)
9
10 # Score CI (using derived formula)
11 score_lower <- (x + z^2/2 - z * sqrt(x + z^2/4)) / n
12 score_upper <- (x + z^2/2 + z * sqrt(x + z^2/4)) / n
13
14 cat("Observed rate:", lambda_hat * 1000, "per 1000 patient-days\n\n")
```

Observed rate: 16 per 1000 patient-days

```
1 cat("Wald 95% CI (per 1000):", round(wald_ci * 1000, 2), "\n")
```

Wald 95% CI (per 1000): 4.91 27.09

```
1 cat("Score 95% CI (per 1000):", round(c(score_lower, score_upper) * 1000, 2))
```

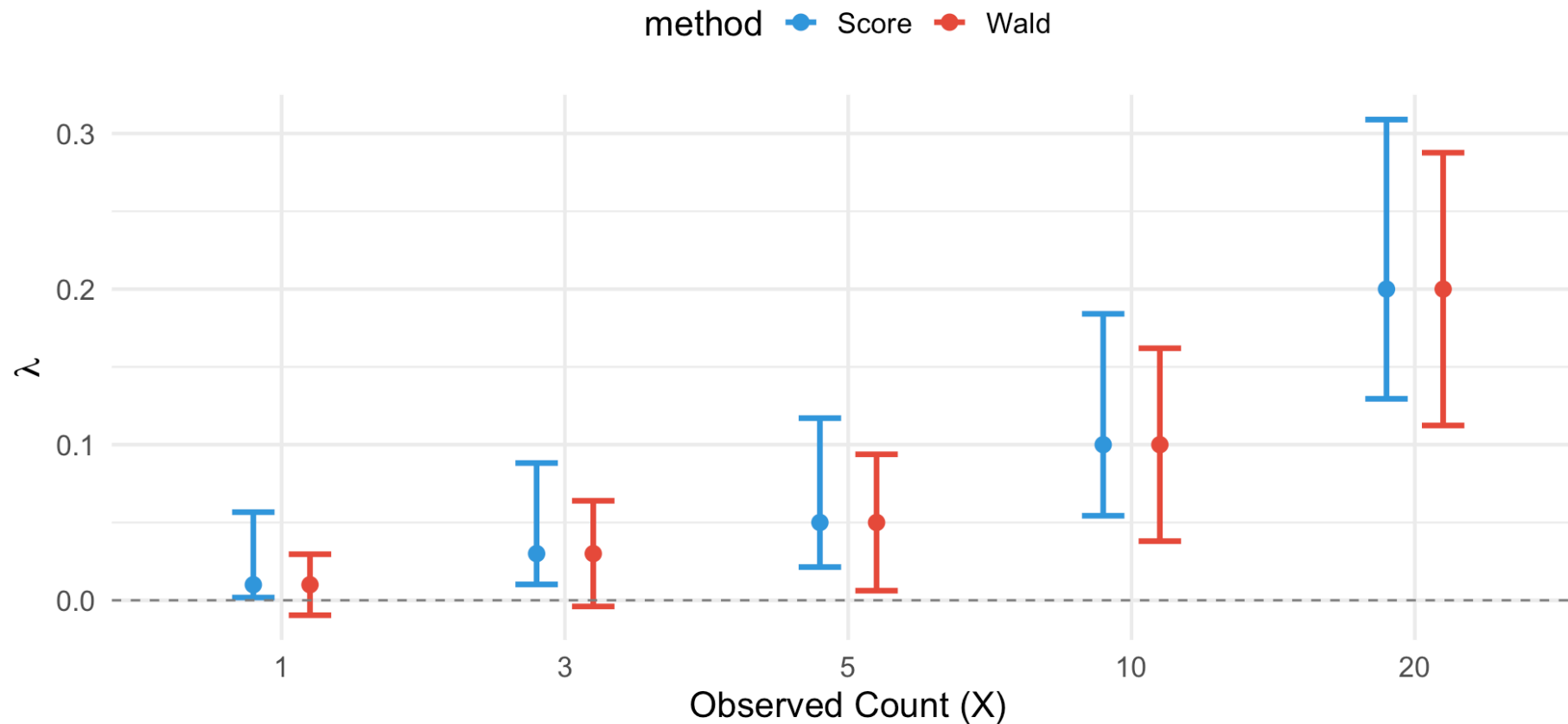
Score 95% CI (per 1000): 8.11 31.58

Comparing Score vs Wald: Small Counts

► Code

Poisson CI Comparison (n = 100 patient-days)

Score intervals avoid negative bounds for small counts



Medical Example: Rare Adverse Events

```
1 # Very rare adverse event: 2 events in 500 patient-years
2 x <- 2; n <- 500
3 z <- qnorm(0.975)
4 lambda_hat <- x / n
5
6 # Wald CI
7 se_wald <- sqrt(lambda_hat / n)
8 wald_ci <- lambda_hat + c(-1, 1) * z * se_wald
9
10 # Score CI
11 score_lower <- (x + z^2/2 - z * sqrt(x + z^2/4)) / n
12 score_upper <- (x + z^2/2 + z * sqrt(x + z^2/4)) / n
13
14 cat("Observed rate:", lambda_hat * 1000, "per 1000 patient-years\n\n")
```

Observed rate: 4 per 1000 patient-years

```
1 cat("Wald 95% CI (per 1000):", round(wald_ci * 1000, 3), "\n")
```

Wald 95% CI (per 1000): -1.544 9.544

```
1 cat(" Problem: Lower bound is NEGATIVE!\n\n")
```

Problem: Lower bound is NEGATIVE!

```
1 cat("Score 95% CI (per 1000):", round(c(score_lower, score_upper) * 1000, 3), "\n")
```

Score 95% CI (per 1000): 1.097 14.586



Exact (Pivotal) Intervals

Score vs Exact Intervals: Key Distinction

 Don't Confuse Score and Exact Intervals!

Score Interval: Based on *asymptotic* normality of the score statistic. Derived by inverting the score test. Approximate.

Exact (Pivotal) Interval: Based on an *exact* distributional result (e.g., chi-squared, t, F). No approximation needed.

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Exact (Pivotal) Interval: Based on an *exact* distributional result (e.g., chi-squared, t, F). No approximation needed.

When both are available, the exact interval is generally preferred!

Exact CI for Exponential Mean

Setting: $X_1, \dots, X_n \sim \text{Exp}(\lambda)$

Pivotal quantity: $2\lambda \sum_{i=1}^n X_i \sim \chi_{2n}^2$ (exact!)

Exact CI for Exponential Mean

Setting: $X_1, \dots, X_n \sim \text{Exp}(\lambda)$

Pivotal quantity: $2\lambda \sum_{i=1}^n X_i \sim \chi_{2n}^2$ (exact!)

Derivation:

$$P\left(\chi_{1-\alpha/2, 2n}^2 < 2\lambda \sum X_i < \chi_{\alpha/2, 2n}^2\right) = 1 - \alpha$$

Exact CI for Exponential Mean

Setting: $X_1, \dots, X_n \sim \text{Exp}(\lambda)$

Pivotal quantity: $2\lambda \sum_{i=1}^n X_i \sim \chi_{2n}^2$ (exact!)

Derivation:

$$P\left(\chi_{1-\alpha/2, 2n}^2 < 2\lambda \sum X_i < \chi_{\alpha/2, 2n}^2\right) = 1 - \alpha$$

Solving for λ :

$$\frac{\chi_{1-\alpha/2, 2n}^2}{2 \sum X_i} < \lambda < \frac{\chi_{\alpha/2, 2n}^2}{2 \sum X_i}$$

Exact CI for Exponential Mean

Setting: $X_1, \dots, X_n \sim \text{Exp}(\lambda)$

Pivotal quantity: $2\lambda \sum_{i=1}^n X_i \sim \chi_{2n}^2$ (exact!)

Derivation:

$$P\left(\chi_{1-\alpha/2, 2n}^2 < 2\lambda \sum X_i < \chi_{\alpha/2, 2n}^2\right) = 1 - \alpha$$

Solving for λ :

$$\frac{\chi_{1-\alpha/2, 2n}^2}{2 \sum X_i} < \lambda < \frac{\chi_{\alpha/2, 2n}^2}{2 \sum X_i}$$

CI for mean $\mu = 1/\lambda$:

$$\frac{2 \sum X_i}{\chi_{\alpha/2, 2n}^2} < \mu < \frac{2 \sum X_i}{\chi_{1-\alpha/2, 2n}^2}$$

Exact vs Wald CI: Exponential Rate

```
1 # Relief time data
2 set.seed(551)
3 relief_times <- rexp(30, rate = 0.5)
4 n <- length(relief_times)
5 sum_x <- sum(relief_times)
6 lambda_hat <- 1 / mean(relief_times)
7 z <- qnorm(0.975)
8
9 # EXACT CI for  $\lambda$  using chi-squared pivotal quantity
10 exact_ci_lambda <- c(
11   qchisq(0.025, df = 2*n) / (2 * sum_x),
12   qchisq(0.975, df = 2*n) / (2 * sum_x)
13 )
14
15 # WALD CI for  $\lambda$  (asymptotic)
16 wald_ci_lambda <- lambda_hat + c(-1, 1) * z * lambda_hat / sqrt(n)
17
18 cat("MLE  $\hat{\lambda}$  =", round(lambda_hat, 3), "\n\n")
```

MLE $\hat{\lambda}$ = 0.498

```
1 cat("Exact ( $\chi^2$ ) CI for  $\lambda$ :", round(exact_ci_lambda, 3), "\n")
```

Exact (χ^2) CI for λ : 0.336 0.692

```
1 cat("Wald CI for  $\lambda$ :", round(wald_ci_lambda, 3), "\n\n")
```

Exact CI for Normal Variance

Setting: $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with μ unknown

Pivotal quantity: $\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$ (exact!)

Exact CI for Normal Variance

Setting: $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with μ unknown

Pivotal quantity: $\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$ (exact!)

Derivation:

$$P \left(\chi_{1-\alpha/2, n-1}^2 < \frac{(n-1)S^2}{\sigma^2} < \chi_{\alpha/2, n-1}^2 \right) = 1 - \alpha$$

Exact CI for Normal Variance

Setting: $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with μ unknown

Pivotal quantity: $\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$ (exact!)

Derivation:

$$P \left(\chi_{1-\alpha/2, n-1}^2 < \frac{(n-1)S^2}{\sigma^2} < \chi_{\alpha/2, n-1}^2 \right) = 1 - \alpha$$

Solving for σ^2 :

$$\frac{(n-1)S^2}{\chi_{\alpha/2, n-1}^2} < \sigma^2 < \frac{(n-1)S^2}{\chi_{1-\alpha/2, n-1}^2}$$

Medical Example: Drug Concentration Variability

```
1 # Simulated concentration data (mg/L)
2 set.seed(551)
3 conc <- rnorm(20, mean = 15, sd = 3)
4
5 n <- length(conc)
6 s2 <- var(conc)
7
8 # EXACT CI for variance (uses chi-squared pivotal quantity)
9 ci_var <- c(
10   (n - 1) * s2 / qchisq(0.975, df = n - 1),
11   (n - 1) * s2 / qchisq(0.025, df = n - 1)
12 )
13
14 # CI for standard deviation
15 ci_sd <- sqrt(ci_var)
16
17 cat("Sample variance:", round(s2, 2), "(mg/L)2\n")
```

Sample variance: 9.5 (mg/L)²

```
1 cat("95% CI for variance:", round(ci_var, 2), "(mg/L)2\n\n")
```

95% CI for variance: 5.5 20.27 (mg/L)²

```
1 cat("Sample SD:", round(sd(conc), 2), "mg/L\n")
```

Bootstrap Review

When to Use Bootstrap

Situation	Best Method
Normal data, σ unknown	t-interval
Non-normal data, large n	t-interval or Bootstrap
Non-normal data, small n	Bootstrap (studentized t)
Complex statistic (median, trimmed mean)	Bootstrap (percentile or studentized)
Ratio or correlation	Bootstrap
Very small n (< 10)	Classical t (bootstrap unreliable)

Bootstrap CI Methods Comparison

Method	Formula	Best When	Coverage
Percentile	$(\hat{\theta}[\alpha/2], \hat{\theta}[1-\alpha/2])$	Symmetric bootstrap dist, low bias	$O(n^{-1/2})$
Simple Bootstrap t	$\hat{\theta} \pm t \cdot SE_{boot}$	Approx normal bootstrap dist	$O(n^{-1/2})$
Studentized Bootstrap t	$\hat{\theta} - t[1-\alpha/2] \cdot SE, \hat{\theta} - t[\alpha/2] \cdot SE$	Skewed data, need accurate coverage	$O(n^{-1})$

New Example: Ratio of Means (Vaccine Efficacy)

In a clinical trial, we want to estimate the **relative risk reduction**:

$$\text{Efficacy} = 1 - \frac{\text{Attack rate in vaccinated}}{\text{Attack rate in placebo}}$$

► Code

Vaccinated: 6/200 (3%)

► Code

Placebo: 12/200 (6%)

► Code

Relative Risk: 0.5

► Code

Vaccine Efficacy: 50 %

Bootstrap CI for Vaccine Efficacy

```
1 # Bootstrap for ratio – need to resample both groups
2 set.seed(551)
3 B <- 10000
4
5 boot_efficacy <- replicate(B, {
6   # Resample cases (treating as binomial outcomes)
7   boot_vax <- rbinom(1, n_vax, cases_vax / n_vax)
8   boot_placebo <- rbinom(1, n_placebo, cases_placebo / n_placebo)
9
10  # Handle zero in denominator
11  if (boot_placebo == 0) return(NA)
12
13  boot_rr <- (boot_vax / n_vax) / (boot_placebo / n_placebo)
14  1 - boot_rr
15 })
16
17 boot_efficacy <- boot_efficacy[!is.na(boot_efficacy)]
18
19 # Percentile CI
20 pct_ci <- quantile(boot_efficacy, c(0.025, 0.975))
21 cat("Bootstrap SE:", round(sd(boot_efficacy) * 100, 1), "%\n")
```

Bootstrap SE: 30.3 %

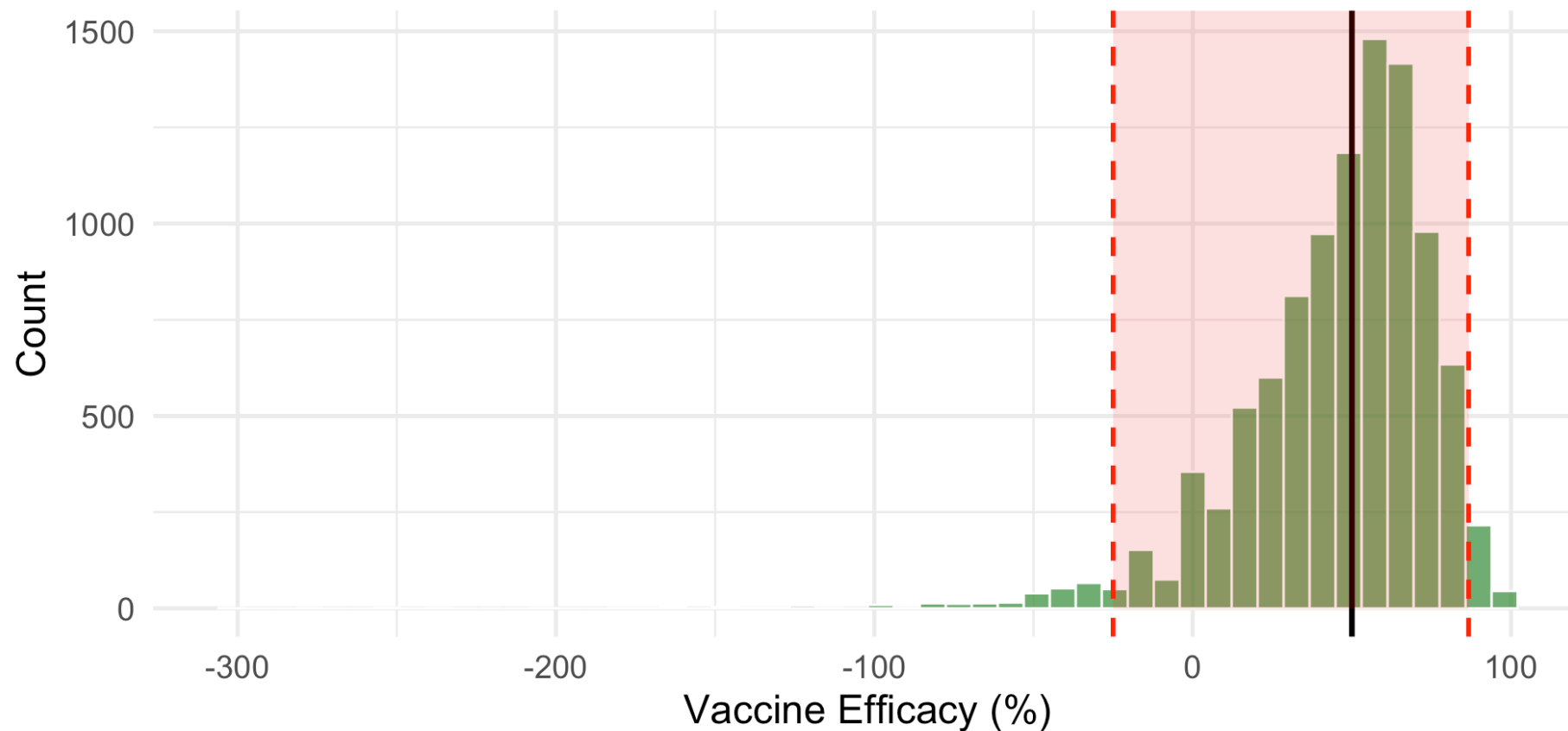


Visualizing Bootstrap Distribution

► Code

Bootstrap Distribution of Vaccine Efficacy

95% CI: (-25%, 86.7%)

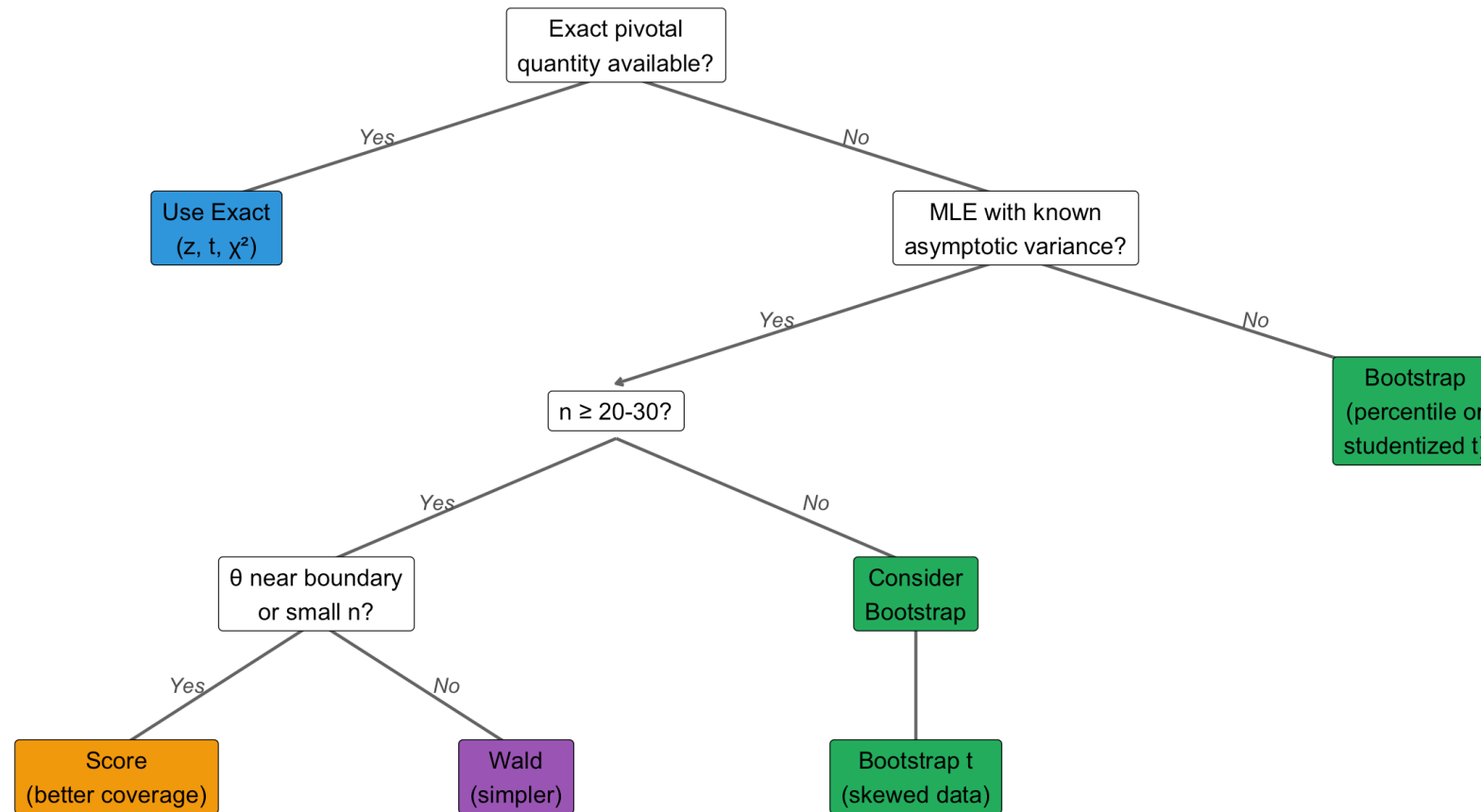


Summary and Decision Guide

CI Method Selection Flowchart

Confidence Interval Method Selection

Score intervals preferred when parameter near boundary (e.g., p close to 0 or 1) or when Wald gives invalid bounds



When to Prefer Each Method

Situation	Recommended Method	Reason
Normal data, σ known	z-interval (exact)	Pivotal quantity available
Normal data, σ unknown	t-interval (exact)	Pivotal quantity available
Proportion, large n, p not extreme	Wald	Simple, adequate coverage
Proportion, small n or p near 0/1	Score (Wilson)	Respects [0,1], better coverage
Poisson rate, small counts	Score	Avoids negative bounds
Exponential mean	Exact (χ^2)	Pivotal quantity exists
Median, trimmed mean, correlation	Bootstrap	No simple SE formula
Ratio of parameters	Bootstrap	Complex sampling distribution

Key Formulas Reference Card

Wald Intervals:

$$\hat{\theta} \pm z_{\alpha/2} \cdot \widehat{SE}(\hat{\theta})$$

Delta Method:

$$SE[g(\hat{\theta})] \approx |g'(\hat{\theta})| \cdot SE(\hat{\theta})$$

Score Principle: Solve $\frac{(\hat{\theta} - \theta)^2}{\text{Var}_{\theta}(\hat{\theta})} = z_{\alpha/2}^2$

Bootstrap Percentile:

$$(\hat{\theta}_{\alpha/2}^*, \hat{\theta}_{1-\alpha/2}^*)$$

Studentized Bootstrap t:

$$\hat{\theta} - t_{1-\alpha/2}^* \cdot SE, \hat{\theta} - t_{\alpha/2}^* \cdot SE$$

where $t_b^* = \frac{\hat{\theta}_b^* - \hat{\theta}}{SE_b^*}$

Looking Ahead

Next Lesson: Hypothesis Testing

- Null and alternative hypotheses
- Type I and Type II errors
- Test statistics and rejection regions
- P-values and their interpretation

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Connection to CIs:

A $100(1 - \alpha)\%$ CI contains exactly those parameter values that would **not** be rejected by a two-sided α -level test.

References

- Devore, Berk, and Carlton. *Modern Mathematical Statistics with Applications* (Springer). Chapter 8.
- Chihara and Hesterberg. *Mathematical Statistics with Resampling and R* (Wiley). Chapters 5, 7.
- Casella and Berger. *Statistical Inference* (2nd ed). Section 10.1 (Delta Method).
- Agresti, A. (2013). *Categorical Data Analysis* (3rd ed). Chapter 1 (Score intervals).

Questions?

Thank you!

Midterm covers Lessons 1-11

Good luck on your exam!